

CHEMICAL CHANGES – pH & NEUTRALISATION

KEY VOCABULARY

pH scale: measures acidity from 0 to 14.

Acid: pH below 7 — source of H^+ ions.

Alkali: a soluble base, pH above 7 — source of OH^- ions.

Neutralisation: $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$.

COMMON MISCONCEPTIONS

~~A high pH means a strong acid.~~

✓ High pH means alkaline; low pH means acidic.

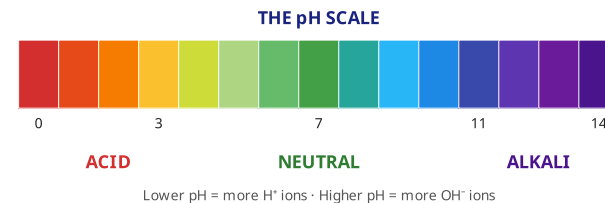
~~All bases are alkalis.~~

✓ Only bases that dissolve in water are alkalis.

~~Neutral means no ions.~~

✓ Neutral (pH 7) has equal H^+ and OH^- .

THE PH SCALE



1 READ THE SCALE

State the pH range of an **acid** and an **alkali**. [2 marks]

2 IONS PRESENT

Name the ion responsible for acidity and the ion for alkalinity. [2 marks]

3 NEUTRALISATION

Write the **ionic equation** for neutralisation. [2 marks]

4 MEASURING PH

Name two ways to **measure the pH** of a solution. [2 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

In neutralisation, ____ ions react with ____ ions to make water.

H^+

OH^-

water

salt

6 APPLY

Suggest why farmers add lime to acidic soil. [1 mark]
